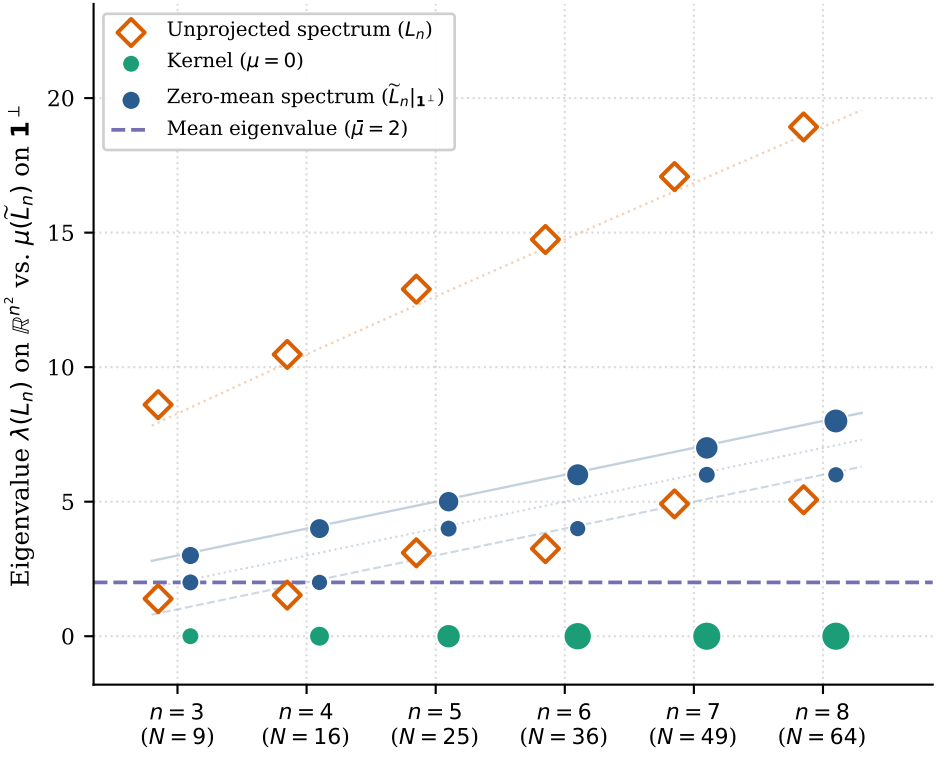
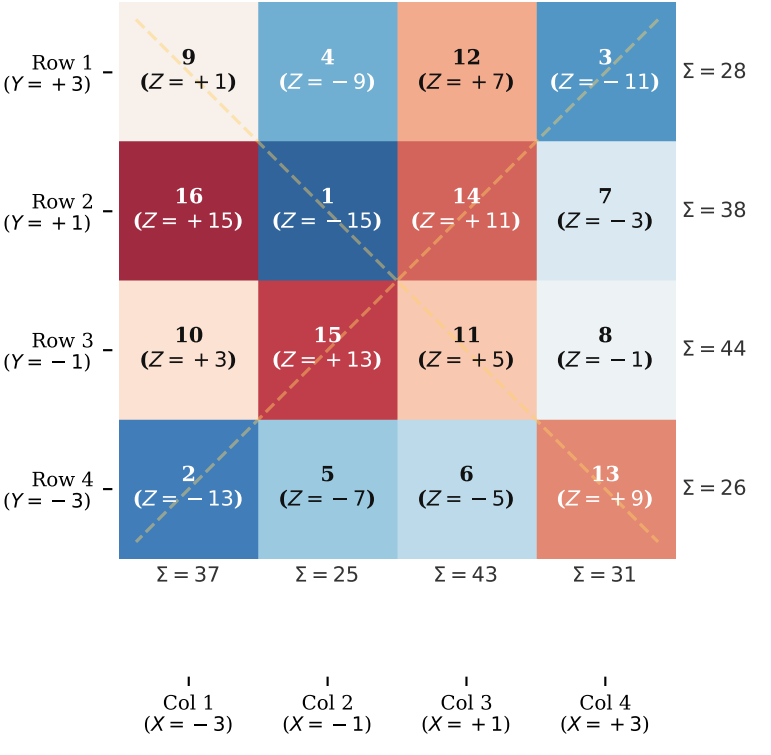

(a) Zero-Mean Integer Spectrum of $\widetilde{L}_n$ on $\mathbf{1}^\perp$
Unprojected spectrum (L_n)
Kernel ($\mu = 0$)
Zero-mean spectrum ($\widetilde{L}_n|_{\mathbf{1}^\perp}$)
Mean eigenvalue ($\bar{\mu} = 2$)
Eigenvalue $\lambda(L_n)$ on $\mathbb{R}^{n^2}$ vs. $\mu(\widetilde{L}_n)$ on $\mathbf{1}^\perp$
0
5
10
15
20
$n = 3$ ($N = 9$)
$n = 4$ ($N = 16$)
$n = 5$ ($N = 25$)
$n = 6$ ($N = 36$)
$n = 7$ ($N = 49$)
$n = 8$ ($N = 64$)


(b) Balanced 4×4 Impostor Square S_{low}
Row 1 ($Y = +3$)
Row 2 ($Y = +1$)
Row 3 ($Y = -1$)
Row 4 ($Y = -3$)
9 ($Z = +1$)
4 ($Z = -9$)
12 ($Z = +7$)
3 ($Z = -11$)
16 ($Z = +15$)
1 ($Z = -15$)
14 ($Z = +11$)
7 ($Z = -3$)
10 ($Z = +3$)
15 ($Z = +13$)
11 ($Z = +5$)
8 ($Z = -1$)
2 ($Z = -13$)
5 ($Z = -7$)
6 ($Z = -5$)
13 ($Z = +9$)
$\Sigma = 28$
$\Sigma = 38$
$\Sigma = 44$
$\Sigma = 26$
$\Sigma = 37$
$\Sigma = 25$
$\Sigma = 43$
$\Sigma = 31$
Col 1 ($X = -3$)
Col 2 ($X = -1$)
Col 3 ($X = +1$)
Col 4 ($X = +3$)